

Uniform Haar noise energy and the exact interacting-law remainder

HOCF–RIESZ, Round 009

18 September 2026

Scope. This synthesis concerns THM–029. Complete proofs, independent reports, qualifications and verification records remain in the immutable Markdown artifacts and canonical ledgers. Its Haar estimates are reference-law estimates. The quantitative transfer to the actual interacting law is still open.

The genuine full inverse and its uniform amplitude

Use the unit Haar torus, $K = -\nabla g$, and the coefficient-one periodic Riesz kernel. Fix $d \geq 3$, $0 < s \leq d - 2$, $T < \infty$, $\nu_* < \infty$, and a smooth real terminal test h . For every $N \geq 2$ and $0 \leq \nu \leq \nu_*$, take external drift zero, reference density one, the actual homogeneous backward Fourier test f , and the unique symmetric terminal-zero full pair inverse Φ . The domain theorem supplies the classical equation off the diagonal,

$$\partial_t \Phi + \nu \Delta_{x,y} \Phi + B\Phi/N + R\Phi = -J, \quad \Phi_T = 0,$$

where

$$Bv = K(x - y) \cdot (\nabla_x - \nabla_y)v, \quad J = K(x - y) \cdot (\nabla f(x) - \nabla f(y)), \quad R = R_x + R_y.$$

It also supplies boundedness, global Haar $H^1 \cap W^{2,1}$, Haar integrability of $\partial_t \Phi$ and $B\Phi$, and the exact particle Itô identity. Its possibly large constants in N will justify operations only; they do not enter the uniform energy constant below.

Set $p = s + 2$ and $a = s/p \in (0, 1)$. The actual Fourier multipliers of f have modulus at most one. Smoothness of h therefore gives uniform bounds on $\nabla f, D^2 f$ over the admitted diffusivity interval. The source satisfies $|J(x, y)| \leq C(1 + |x - y|^{-s})$, so $\sup_t \|J_t\|_1 \leq C_J$ independently of N, ν . The full earlier radial comparison gives the positive profile

$$F_{N,\tau}(r) = \frac{N}{4} \left[(r^p + 2sp\tau/N)^{2/p} - r^2 \right] \leq \frac{(2sp)^{2/p}}{4} N^a \tau^{2/p}.$$

The local cutoff and its full diffusion cross term have fixed annular bounds. Consequently the genuine base-pair source potential satisfies $\sup_t \|U_N(t)\|_\infty \leq C_U N^a$. The exact bounded-response Volterra series, with both response slots, yields

$$\sup_t \|\Phi_t\|_\infty \leq C_\infty N^a, \quad C_\infty = C_U e^{2\|D\|_{\text{TV}} T}, \quad D = \text{div } K.$$

All displayed constants depend only on d, s, T, ν_*, h and the fixed kernel and cutoffs. No weighted derivative constant is hidden here.

Energy integration retains the Coulomb flux

Write $\psi(z, y) = \Phi(z + y, y)$. Then $B = 2K(z) \cdot \nabla_z - K(z) \cdot \nabla_y$. Delete the relative tube $|z| \leq \epsilon$. The diffusion boundary term tends to zero because Φ is bounded and $|\nabla \Phi| \leq C_N |z|^{-q_1}$ for an allowed $1 < q_1 < d/2 < d - 1$. Its size is at most $C_N \epsilon^{d-1-q_1}$. Global H^1 and $W^{2,1}$ then give

$$\int \Phi \Delta_{x,y} \Phi = -\|\nabla_{x,y} \Phi\|_2^2.$$

The exact internal-drift formula is

$$\int_{|z|>\epsilon} \Phi B \Phi = - \int_{|z|=\epsilon} \int K(z) \cdot \frac{z}{\epsilon} \psi(z, y)^2 dy dS_z - \int_{|z|>\epsilon} (\operatorname{div}_{\text{cl}} K)(z) \psi^2 dz dy.$$

Near zero, $K(z) = s|z|^{-s-2}z + k(z)$ with $|k(z)| \leq L_k|z|$. Thus the surface term is nonpositive for small ϵ . Below Coulomb it vanishes. At Coulomb its sign is retained; no diagonal trace or vanishing flux is assumed. Since $\operatorname{div}_{\text{cl}} K \geq -\kappa$ and $B\Phi \in L^1$, passage to the limit gives

$$\int \Phi B \Phi \leq \kappa \|\Phi\|_2^2.$$

This is an upper bound, not nonpositivity of the full internal drift. A constant kernel at Coulomb has cancelling bulk and boundary terms.

The two homogeneous responses are minus convolution with the even finite measure D in the respective slots. Its exact Fourier coefficients are

$$d_0 = 0, \quad d_k = 4\pi^2 c_{d,s} |k|^{s+2-d} \geq 0 \quad (k \neq 0).$$

They are bounded in this exponent range. Fourier polynomial approximation and the finite-measure L^2 operator bound prove

$$\langle \Phi, R\Phi \rangle = - \sum_{k,\ell} (d_k + d_\ell) |\widehat{\Phi}(k, \ell)|^2 \leq 0.$$

This sign belongs to the homogeneous convolution responses.

Multiply the classical equation by Φ and integrate. The fixed- N weighted time-derivative majorant justifies the energy derivative. The terminal-zero convention gives

$$\nu \int_0^T \|\nabla_{x,y} \Phi_t\|_2^2 dt + \frac{1}{2} \|\Phi_0\|_2^2 = \int_0^T [\langle \Phi, J \rangle + N^{-1} \langle \Phi, B\Phi \rangle + \langle \Phi, R\Phi \rangle] dt.$$

Therefore, using $2a - 1 \leq a$ and unit Haar mass,

$$\boxed{\nu \int_0^T \|\nabla_{x,y} \Phi_t\|_2^2 dt \leq T(C_J C_\infty + \kappa C_\infty^2) N^a.}$$

The fixed- N domain constants have disappeared. At zero noise the left side is zero; no division by ν or uniform unweighted gradient estimate follows.

Exact Haar noise factors

Put $G(x, y) = \nabla_x \Phi(x, y)$, $A(x) = \int G(x, y) dy$, and $H = G - A$. For the mean-field-centered, ordered deleted-pair statistic,

$$P[\Phi] = \frac{1}{2N^2} \sum_{i \neq j} \Phi(X_i, X_j) - \frac{1}{N} \sum_i \int \Phi(X_i, y) dy + \frac{1}{2} \int \Phi dx dy,$$

its exact particle gradient is

$$\nabla_i P = N^{-2} \left[\sum_{j \neq i} H(X_i, X_j) - A(X_i) \right].$$

Under product Haar measure m_N , the H terms have zero conditional means given X_i . Squaring before integrating and retaining the deleted-label background term gives

$$\boxed{\int \sum_i |\nabla_i P|^2 dm_N = \frac{N-1}{N^3} \|G\|_2^2 - \frac{N-2}{N^3} \|A\|_2^2.}$$

At $N = 2$ the second coefficient is zero; an additive pair kernel still leaves $N^{-3}\|A\|_2^2$. No diagonal is inserted to change the normalization.

For positive $\nu = 1/\beta_N$, let $b_N = \min(\beta_N, 1)$ and $\sigma_N^2 = Nb_N$. Since pair symmetry gives $\|\nabla_{x,y}\Phi\|_2^2 = 2\|G\|_2^2$, the reference noise functional obeys

$$Q_P^{\text{Haar}} := \sigma_N^2 \int_0^T \int 2\nu \sum_i |\nabla_i P_t|^2 dm_N dt \leq Cb_N N^{-2/(s+2)}.$$

For $\eta_N(f_t)$, the corresponding leading reference functional is $Q_f^{\text{Haar}} = 2\nu b_N \int_0^T \|\nabla f_t\|_2^2 dt \leq C$. Cauchy–Schwarz in particle components, Haar variables and time proves

$$\sigma_N^2 \int_0^T \int \left| 2\nu \sum_i N^{-1} \nabla f_t(X_i) \cdot \nabla_i P_t \right| dm_N dt \leq \sqrt{Q_f^{\text{Haar}} Q_P^{\text{Haar}}} \leq C\sqrt{b_N} N^{-1/(s+2)}.$$

The absolute value remains inside both integrals. These statements assume bounded diffusivity; general subcritical sequences with $\beta_N \rightarrow 0$ are not covered. Zero-noise functionals vanish separately.

The actual-law estimate still required

For an exchangeable law with the stated products integrable, the same literal square yields

$$\begin{aligned} \mathbb{E} \sum_i |\nabla_i P|^2 &= N^{-3} [(N-1)\mathbb{E}|H_{12}|^2 + (N-1)(N-2)\mathbb{E}(H_{12} \cdot H_{13}) \\ &\quad - 2(N-1)\mathbb{E}(H_{12} \cdot A_1) + \mathbb{E}|A_1|^2]. \end{aligned}$$

At $N = 2$ the triple term is absent; no third coordinate is defined. The dynamics commute with label permutations and common translations. Pathwise uniqueness therefore shows that their laws from iid Haar preparation are exchangeable and have Haar one-body marginals. They do not establish product higher marginals. Let $F_t^{(2)}, F_t^{(3)}$ be their pair and triple densities, and define

$$\begin{aligned} \delta_2(t) &= \int |H_t(x, y)|^2 (F_t^{(2)} - 1) dx dy, \\ \delta_3(t) &= \int H_t(x, y) \cdot H_t(x, z) (F_t^{(3)} - 1) dx dy dz, \\ \delta_A(t) &= \int H_t(x, y) \cdot A_t(x) (F_t^{(2)} - 1) dx dy. \end{aligned}$$

All exist at fixed N by the established density domination and Haar H^1 . The actual expected scaled corrector bracket satisfies the exact reduction

$$Q_P^{\text{actual}} - Q_P^{\text{Haar}} = \frac{2\nu b_N}{N^2} \int_0^T [(N-1)\delta_2 + (N-1)(N-2)\delta_3 - 2(N-1)\delta_A] dt.$$

A vanishing bound for this signed combination, or a suitable upper bound for its positive part, would suffice. Replacing each deviation by its absolute value gives a stronger sufficient condition, not a necessary one. The available full-density comparison only gives $Q_P^{\text{actual}} \leq e^{(N-1)\kappa T} Q_P^{\text{Haar}}$, which need not vanish. A uniform estimate for the displayed marginal remainder is open.

The leading actual bracket expectation already equals its Haar value because it depends only on the one-body marginal. An actual corrector self-bracket bound would therefore imply the corresponding expected absolute cross bound. The evolved cubic residual and its lower contractions still need their own estimates; no fluctuation theorem or critical hierarchy closure follows. The old energy-floor condition $\beta_N N^{2s/d-1} \rightarrow 0$, full microscopic subcriticality $\lambda_N = \beta_N N^{s/d-1} \rightarrow 0$, and microscopic criticality $\lambda_N \rightarrow \lambda \in (0, \infty)$ remain distinct.